

Education

- 2021-2025 **PhD in Computer Science**, UNICAMP, Brazil. Thesis: Deep Learning.
- 2019-2021 **M.Sc. in Computer Science**, UNICAMP, Brazil. Thesis: Conversational Systems.
- 2012-2017 **B.Sc./Eng. in Computer and Systems Engineering**, UNSAAC, Peru. Achieved a ranking in the upper fifth percentile throughout the course of my academic years. Thesis: Computer Vision.

Awards and Recognitions

- 2021 **Shell Oil Company**, Industry Research Scholarship.
- 2019 **Sinch Latin America**, Industry Research Scholarship.
- 2018 **First place** in the AgroHack hackathon. Developed a mobile app for the recognition, monitoring, and support of plant diseases using IBM Cloud services. Cusco, Peru.
- 2014 **UNSAAC**, Winner of the Technology Fair UNSAAC with a project using PIC 16F84A.

Languages and Technologies

Machine/Deep Learning (ML/DL), Natural Language Processing (NLP), Computer Vision (CV), Text-To-Speech Processing (TTS), OpenCV, Scikit-Learn, Matplotlib, Pandas, Keras, PyTorch, TensorFlow, HuggingFace, LLM.

Projects, Experience, and Publications

- 2025-Present **Postdoctoral Researcher, UNICAMP (Sponsored by Shell Oil). Advisor: Anderson Rocha**
- Developed ML-based frameworks for automated interpretation of large-scale 3D/4D seismic data.
 - Applied supervised/unsupervised learning to detect reservoir dynamics using dRMS and dAmp attributes.
 - Designed QC and confidence-estimation methods to improve reliability of time-lapse seismic signals.
 - Bridged machine learning outputs with reservoir engineering decision-making.
- 2024-Present **Software Developer in TTS, Samsung Instituto De Desenvolvimento (SiDi), Brazil.**
- Collaborated with global teams to integrate Bixby and Samsung Interpreter into Samsung's ecosystem, ensuring support for Brazilian Portuguese, European Portuguese, and Mexican Spanish across devices in over 200 countries.
 - Enhanced the TTS model for Bixby and Samsung Interpreter by curating, creating, and reviewing audio data, adjusting training hyperparameters, and validating outputs to reduce prosody errors, mispronunciations, and a robotic voice.
- 2021-2025 **DL Researcher, UNICAMP (Sponsored by Shell Oil). Advisor: Anderson Rocha**
- Developed an efficient end-to-end proxy to replace numeric simulators during data assimilation. This approach tackled high dimensionality, limited sample sizes, and the need for accurate post-dimensionality reduction reconstruction: **Paper1, Paper2, Paper3, Paper4, Paper5.**
 - Implemented a 3D CNN to achieve dimensionality reduction and replicate the simulator response. Augmented the dataset with samples generated by GANs. Utilized a VAE to reconstruct each 3D geological uncertainty post-dimensionality reduction.
 - Achieved significant improvements: reduced simulation times by 52.45% while maintaining high accuracy. The replication of the simulation response incurred an error of only 3.4%, and the reconstruction process exhibited a 99% correlation between the original and reconstructed 3D uncertainties. This innovation streamlined data assimilation, enhancing efficiency in geological modeling and decision-making.
- 2021 **ML/DL Instructor, University of Engineering and Technology (UTEC), Lima, Peru.**
- Delivered a comprehensive ML/DL course integrating theory and applied projects.
- 2019 - 2021 **NLP Researcher, UNICAMP (Sponsored by Sinch). Advisor: Julio Cesar dos Reis**
- Worked on enhancing machine learning and NLP models at Sinch Latin America by developing techniques to reduce labeled data requirements and optimize context management in dialog flows: **Paper6, Paper7.**
 - Proposed an active learning-based technique to minimize labeled data using a CNN architecture and developed a windows-based approach for managing context, focusing on previous utterances.
 - Conducted a comparative analysis between a pretrained BERT model in Portuguese and the proposed methodology.
 - Reduced time and data labeling costs, achieving a 2.5% accuracy improvement in intent classification.
- 2017 - 2018 **CV Researcher, UNSAAC, Cusco, Peru. Advisor: Lauro Enciso Rodas**
- Worked on recognizing images of historical buildings in Cusco by developing a recognition system and comparing the effectiveness of bag-of-words and CNN methods: **Paper8, Paper9.**
 - Implemented image recognition using OpenCV, Scikit-learn, and TensorFlow.
 - Enhanced recognition accuracy and efficiency through the application of CNNs in contrast to traditional methods
- 2016 - 2018 **Software Engineer, Money Express Center S.A.C., Cusco, Peru.**
- Developed a new monetary transactions system to improve efficiency, handling both desktop and web applications with a focus on transaction speed and customer service.
 - Created the system using Java and Node.js, implementing real-time transaction visualization with Socket technologies.
 - Achieved a 15% reduction in customer service time by providing instant transaction updates.

Community Service	
2023-Present	Volunteer, IEEE INTERCON Reviewer.
2021	Volunteer, Data Science Woman Peru.
2012-2016	Volunteer, ACM chapter Cusco.

Languages	
Spanish	Mother tongue.
Portuguese	Fluent.
English	Advanced.